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**Java SE 27 & JDK 27**

# Class MemoryUsage

java.lang.Object
    java.lang.management.MemoryUsage

---

```
public class MemoryUsage
extends Object
```

A `MemoryUsage` object represents a snapshot of memory usage. Instances of the `MemoryUsage` class are usually constructed by methods that are used to obtain memory usage information about individual memory pool of the Java virtual machine or the heap or non-heap memory of the Java virtual machine as a whole.

A `MemoryUsage` object contains four values:

| Value | Description |
|---|---|
| `init` | represents the initial amount of memory (in bytes) that the Java virtual machine requests from the operating system for memory management during startup. The Java virtual machine may request additional memory from the operating system and may also release memory to the system over time. The value of `init` may be undefined. |
| `used` | represents the amount of memory currently used (in bytes). |
| `committed` | represents the amount of memory (in bytes) that is guaranteed to be available for use by the Java virtual machine. The amount of committed memory may change over time (increase or decrease). The Java virtual machine may release memory to the system and `committed` could be less than `init`. `committed` will always be greater than or equal to `used`. |
| `max` | represents the maximum amount of memory (in bytes) that can be used for memory management. Its value may be undefined. The maximum amount of memory may change over time if defined. The amount of used and committed memory will always be less than or equal to `max` if `max` is defined. A memory allocation may fail if it attempts to increase the used memory such that `used > committed` even if `used <= max` would still be true (for example, when the system is low on virtual memory). |

Below is a picture showing an example of a memory pool:

```
        |--------|
           init
        |---------------|
               used
        |--------------------------|
                  committed
        |-------------------------------------------|
                             max
```

## MXBean Mapping

`MemoryUsage` is mapped to a `CompositeData` with attributes as specified in the `from` method.

**Since:**

1.5

## *Constructor Summary*

**Constructors**

| Constructor | Description |
|---|---|
| `MemoryUsage(long init, long used, long committed, long max)` | Constructs a `MemoryUsage` object. |

## *Method Summary*

**All Methods** | **Static Methods**

| Modifier and Type | Method | Description |
|---|---|---|
| `static MemoryUsage` | `from(CompositeData cd)` | Returns a `MemoryUsage` object represented by the given `CompositeData`. |
| `long` | `getCommitted()` | Returns the amount of memory in bytes that is committed for the Java virtual machine to use. |
| `long` | `getInit()` | Returns the amount of memory in bytes that the Java virtual machine initially requests from |

| | | |
|---|---|---|
| | | the operating system for memory management. |
| long | **getMax**() | Returns the maximum amount of memory in bytes that can be used for memory management. |
| long | **getUsed**() | Returns the amount of used memory in bytes. |
| String | **toString**() | Returns a descriptive representation of this memory usage. |

**Methods declared in class Object**

clone, equals, finalize, getClass, hashCode, notify, notifyAll, wait, wait, wait

## *Constructor Details*

### MemoryUsage

```
public MemoryUsage(long init,
                   long used,
                   long committed,
                   long max)
```

Constructs a `MemoryUsage` object.

**Parameters:**

`init` - the initial amount of memory in bytes that the Java virtual machine allocates; or `-1` if undefined.

`used` - the amount of used memory in bytes.

`committed` - the amount of committed memory in bytes.

`max` - the maximum amount of memory in bytes that can be used; or `-1` if undefined.

**Throws:**

`IllegalArgumentException` - if

- the value of `init` or `max` is negative but not `-1`; or
- the value of `used` or `committed` is negative; or
- `used` is greater than the value of `committed`; or
- `committed` is greater than the value of `max` `max` if defined.

## *Method Details*

### getInit

```
public long getInit()
```

Returns the amount of memory in bytes that the Java virtual machine initially requests from the operating system for memory management. This method returns `-1` if the initial memory size is undefined.

**Returns:**

the initial size of memory in bytes; `-1` if undefined.

### getUsed

```
public long getUsed()
```

Returns the amount of used memory in bytes.

**Returns:**

the amount of used memory in bytes.

### getCommitted

```
public long getCommitted()
```

Returns the amount of memory in bytes that is committed for the Java virtual machine to use. This amount of memory is guaranteed for the Java virtual machine to use.

**Returns:**

the amount of committed memory in bytes.

### getMax

```
public long getMax()
```

Returns the maximum amount of memory in bytes that can be used for memory management. This method returns `-1` if the maximum memory size is undefined.

This amount of memory is not guaranteed to be available for memory management if it is greater than the amount of committed memory. The Java virtual machine may fail to allocate memory even if the amount of used memory does not exceed this maximum size.

**Returns:**

the maximum amount of memory in bytes; `-1` if undefined.

## toString

```
public String toString()
```

Returns a descriptive representation of this memory usage.

**Overrides:**

`toString` in class `Object`

**Returns:**

a string representation of the object

## from

```
public static MemoryUsage from(CompositeData cd)
```

Returns a `MemoryUsage` object represented by the given `CompositeData`. The given `CompositeData` must contain the following attributes:

| Attribute Name | Type |
|---|---|
| init | `java.lang.Long` |
| used | `java.lang.Long` |
| committed | `java.lang.Long` |
| max | `java.lang.Long` |

**Parameters:**

`cd` - `CompositeData` representing a `MemoryUsage`

**Returns:**

a `MemoryUsage` object represented by `cd` if `cd` is not `null`; `null` otherwise.

**Throws:**

`IllegalArgumentException` - if `cd` does not represent a `MemoryUsage` with the attributes described above.

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Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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